From Per-Pass Savings to End-to-End Energy: A Pre-Registered Study of Forward-Only Learning

Przemysław Spyra[a,*], Witold Dzwinel[a]

[a] *AGH University of Krakow Faculty of Computer Science al. ˜Mickiewicza 30 30-059 Krak\ 'ow Poland*

Abstract

Backpropagation-free training is often described as energy-efficient because it removes the backward sweep, but a per-pass saving does not imply an end-to-end saving: a method that needs more passes can consume more energy overall. This paper separates the two effects. We define the per-pass energy advantage a and the pass multiplier m, derive the condition $a > m$ under which a forward-only method wins end to end, and test that condition in a pre-registered, hardware-instrumented family of experiments on NVIDIA A100 GPUs. The design covers five training arms, two image-classification datasets, twenty paired seeds, tuned BP controls, a scheduled-BP control, and budget-matched controls. [W1 PLACEHOLDER: insert the W1 end-to-end result and its pre-registered uncertainty statement here; do not insert a number before the frozen analysis is complete.] The contribution is a measurement framework and a confirmatory test, not a FLOP-count argument.

Keywords: forward-only learning, Mono-Forward, backpropagation, energy measurement, pre-registration, reproducibility

1. Introduction

Deep neural networks are trained overwhelmingly with backpropagation (BP) [**?**]. BP is effective, but it couples the layers: the backward sweep must retain activations and must wait for the layer above. Forward-only algorithms replace the global backward sweep with local learning signals.

*Corresponding author.

Email address: pspyra@agh.edu.pl (Przemysław Spyra)

Hinton's Forward-Forward (FF) [**?**], Cascaded Forward (CaFo) [**?**], and Mono-Forward (MF) [**?**] are the three representatives considered here. Their appeal is architectural as well as numerical: no global backward graph, no activation store for a backward sweep, and a natural mapping to pipelined or memory-constrained hardware.

The efficiency case is usually made with operation counts. That is not enough. A method can execute fewer FLOPs and still consume more energy if it runs longer, uses the device poorly, or moves more data. The PPAM 2026 paper measured energy on hardware and found that removing the backward pass is not by itself a saving: FF and CaFo consumed more energy than a tuned BP baseline in the tested configurations, while MF was the exception in the matched-data-pass protocol. The same study also found that MF needed more passes to reach its own convergence, which is exactly the regime in which a per-pass advantage can disappear end to end.

This paper extends that work in four ways. First, it makes the end-to-end argument explicit: Section 4 defines per-pass energy, pass multiplier, and the win condition, including fixed-cost and uncertainty corrections. Second, it reports a pre-registered confirmatory family (W1) with five arms, two datasets, twenty paired seeds, tuned BP controls, a scheduled-BP control, and budget-matched controls; the protocol was frozen before any W1 outcome was analysed. Third, it adds a measurement protocol that separates matched-data-pass energy from matched-accuracy energy, so that a per-pass result cannot be presented as an end-to-end result. Fourth, it releases the frozen configurations, analysis code, and table shells needed to reproduce the comparison.

Incremental contribution. This manuscript is an expanded version of the PPAM 2026 paper. The conference paper established the hardware measurement and the matched-data-pass comparison. The journal version adds the formal cost model (Section 4), the pre-registered W1 design (Section 7), the end-to-end and control analyses (Section 8), the budget-matched Protocol B analysis, and the extended discussion of validity. The exact incremental statement will be repeated in the cover letter at submission.

2. Background and Related Work

2.1. Backpropagation and its cost

BP computes a global gradient by a forward sweep followed by a backward sweep. The backward sweep requires the activation of each layer to remain available until its gradient has been computed. The resulting memory and

synchronisation costs are well known; the energy cost is less often measured. [W1 PLACEHOLDER: cite the specific BP energy literature used in the final related-work section; do not reuse the PPAM bibliography without checking each entry.]

2.2. Forward-only algorithms

FF replaces the global objective with layer-local goodness objectives. CaFo cascades local classifiers and trains them sequentially. MF combines a forward stage with an auxiliary objective and a detached path between stages. The three methods are not interchangeable: they differ in architecture, parameter count, local objective, and the number of forward stages per data pass. A fair comparison therefore has to fix the data-pass definition and the tuning budget, not only the nominal FLOP count.

2.3. Energy measurement

Hardware energy is the integral of instantaneous power over time. NVML provides device-level power samples at a fixed interval; the integral is an estimate whose error depends on the sampling interval, the clock behaviour, and the treatment of idle periods. The W1 traces retain the raw samples and the segment boundaries so that the estimator can be audited. [W1 PLACEHOLDER: add the measurement-error discussion and the exact NVML version used in the final runs.]

2.4. Fair tuning and pre-registration

A comparison between a tuned baseline and a default-configured alternative is not a comparison of algorithms. The W1 design freezes the search space, the trial budget, the sampler, and the stopping rule before the confirmatory runs. The BP-bestval control gives BP the best validation-selected configuration from its own search; the BP-scheduled control adds a cosine schedule and disables early stopping; the budget-matched controls equalise the pass budget using the frozen Protocol B rule. These controls address different objections and are reported separately.

3. End-to-End Energy of Forward-Only Training

4. End-to-End Energy of Forward-Only Training

This section separates the two quantities that are often conflated in discussions of backpropagation-free learning: the energy of one training pass

and the energy needed to reach a specified accuracy. The distinction is elementary, but it is the difference between a per-pass efficiency claim and an end-to-end efficiency claim. The W1 family is designed to measure both quantities under one instrumentation boundary.

Status labels. A statement labelled **[Definition]** fixes a quantity or a measurement boundary; **[Derivation]** means the statement follows algebraically from the definitions; **[Conjecture]** means the statement is an empirical hypothesis that the W1 design can test only in part. Unlabelled sentences are connective exposition and make no independent empirical claim. No W1 outcome value appears in this section.

4.1. Definitions and measurement boundary

[Definition] Fix a training set $\mathcal{D}$, a model, an optimisation algorithm, and an evaluation metric A (here, test accuracy). Let $A^\star$ be a target accuracy fixed before the comparison. A *data pass* is one traversal of $\mathcal{D}$ through the training computation that the algorithm actually performs. For backpropagation (BP), one data pass comprises the forward sweep, the backward sweep, and the parameter update. For a forward-only algorithm, one data pass comprises every forward stage, local loss, and local update that the algorithm performs during that traversal. This definition deliberately counts the algorithm as implemented, not an idealised forward FLOP count.

[Definition] Let e_f and e_b be the mean energy consumed per data pass by the forward-only method and by BP, respectively, measured at the same device, software stack, batch size, and instrumentation boundary. Let $N_f(A^\star)$ and $N_b(A^\star)$ be the number of data passes each method needs to reach $A^\star$ for the first time. When the target is unambiguous we write N_f and N_b.

[Definition] The *per-pass energy advantage* of the forward-only method is

$$a \;=\; \frac{e_b}{e_f}, \tag{1}$$

and its *pass multiplier* is

$$m \;=\; \frac{N_f}{N_b}. \tag{2}$$

Both quantities are dimensionless. $a > 1$ means that one forward-only pass is cheaper than one BP pass; $m > 1$ means that the forward-only method needs more passes to reach the same target. Neither statement alone determines the end-to-end comparison.

[Definition] Ignoring fixed start-up and evaluation costs for the moment, the end-to-end training energies are

$$E_f \;=\; N_f\, e_f, \qquad E_b \;=\; N_b\, e_b. \tag{3}$$

Equations (3) are the formal content of the phrase "end-to-end energy is per-pass energy times the number of passes". The per-pass energies are not assumed equal, and the pass counts are not assumed equal.

4.2. The win condition

[Derivation] From (1)–(3), the ratio of end-to-end energies is

$$R \;=\; \frac{E_f}{E_b} \;=\; \frac{N_f}{N_b}\,\frac{e_f}{e_b} \;=\; \frac{m}{a}. \tag{4}$$

Therefore, in the no-overhead model,

$$E_f < E_b \quad\Longleftrightarrow\quad a > m. \tag{5}$$

A forward-only method with a per-pass advantage a wins end to end exactly when its pass multiplier is smaller than that advantage. If $a = m$, the two methods have equal training energy in this model; if $a < m$, the forward-only method is cheaper per pass and more expensive end to end.

[Derivation] The result is not a statement about FLOPs. The quantities e_f and e_b are measured energies, so (5) already includes power draw, achieved clock, memory traffic, and any idle time inside the pass. A method can execute fewer arithmetic operations and still have a smaller a if it utilises the device poorly or moves more data.

[Derivation] Fixed costs change the threshold. Let

$$E_f = c_f + N_f e_f, \qquad E_b = c_b + N_b e_b, \tag{6}$$

where c_f and c_b collect start-up, data-loading, evaluation, and other run-level costs. Substituting (1)–(2) and rearranging gives

$$a \;>\; \frac{m}{1 + \dfrac{c_b - c_f}{N_b e_b}}. \tag{7}$$

When $c_f = c_b$, (7) reduces to (5). A forward-only method with larger fixed costs must clear a higher per-pass advantage; a method with smaller fixed costs

can tolerate a larger pass multiplier. The sign of $c_b - c_f$ therefore matters, and the W1 traces retain the run boundary and the recorded evaluation timestamps; where a fixed cost cannot be separated, it is bounded rather than assumed zero.

[Definition] The *matched-accuracy end-to-end energy* of a method is the cumulative energy consumed from the start of training until the first pass at which the evaluation metric reaches $A^\star$. This is the quantity compared in (3). If a method never reaches $A^\star$, it has no finite matched-accuracy energy and is reported as a censored run rather than silently assigned the energy of its last pass.

[Conjecture] Removing the backward sweep is not by itself sufficient for an end-to-end gain. The empirically relevant question is whether the measured pair (a, m) satisfies (5); a method that is cheaper per pass can still lose if its convergence is slow enough. This conjecture is the organising hypothesis of the W1 family, not a result of this section.

4.3. What the W1 design can test

The cost model is useful only if its quantities are measurable. Table 1 lists the assumptions behind (5)–(7) and the part of the W1 design that can test each one. The table is deliberately explicit about what is direct evidence, what is partial evidence, and what is outside the fixed W1 architecture.

[Definition] The *measured* per-pass advantage is $\hat{a} = \bar{e}_b/\bar{e}_f$, where the bars denote the protocol-level means over the pre-registered seeds. The *measured* pass multiplier is $\hat{m} = \bar{N}_f/\bar{N}_b$ at the same target. These are estimators of (1) and (2), not definitions of new quantities. Their uncertainty must be propagated: a point estimate $\hat{a} > \hat{m}$ is not by itself evidence that $a > m$ in the population of seeds, because both quantities are measured with run-to-run variation.

[Derivation] If $\hat{a}$ and $\hat{m}$ are approximately independent with relative standard errors s_a and s_m, the relative standard error of $\hat{R} = \hat{m}/\hat{a}$ is approximately $\sqrt{s_a^2 + s_m^2}$. A decision rule based on (5) therefore needs a confidence interval for R, not only a comparison of two point estimates. The W1 analysis reports the interval and the paired-seed structure; the theory section does not assume that the interval excludes one.

[Conjecture] The pair (a, m) may be architecture-dependent in a way that the fixed W1 MLP cannot reveal. In particular, a may improve with width because activation memory and parameter traffic grow differently in the two methods, while m may improve with a better local objective or a

Table 1: Assumptions of the end-to-end cost model and their W1 tests. "Direct" means the W1 instrumentation records the required quantity; "partial" means the design constrains the assumption but does not vary it; "not tested" means the assumption is a scope limitation. No outcome value is shown.

Assumption	W1 evidence	Comment
e_f, e_b measured at one boundary	direct	NVML energy integration with the same start/stop rule for every arm.
Same target $A^\star$	direct	Pre-registered target and evaluation protocol; censored runs reported.
N_f, N_b counted as implemented	direct	Epoch/pass counters; no FLOP-based substitution.
Per-pass energy stable over training	partial	Per-stage sums versus total energy; drift check across seeds.
Fixed costs estimated	partial	Run boundary and evaluation timestamps; fixed costs bounded where not separable.
Same data pipeline and batch size	direct (design)	Frozen configs; no per-arm preprocessing.
Same hardware generation	partial	A100 partition, but node-to-node variation is recorded, not eliminated.
Scaling to larger widths/depths	not tested	W1 fixes the 3×2000 MLP; scale is a separate conjecture.

schedule that stops earlier. A scale-dependent crossover, if it exists, would require a separate width/depth study; W1 can falsify the claim that backward removal is sufficient at its fixed scale, but it cannot establish a general scaling law.

4.4. Interpretation rules

[Definition] We call a result *end-to-end favourable* to the forward-only method when the matched-accuracy energy ratio satisfies $R < 1$ with a pre-registered uncertainty rule. We call it *per-pass favourable* when $a > 1$ but make no end-to-end claim from that fact alone. This terminology prevents the common slide from "cheaper per pass" to "cheaper overall".

[Derivation] Three boundary cases follow directly from (4). First, if $a \leq 1$, the forward-only method cannot win end to end at any $m \geq 1$ in the

no-overhead model. Second, if $a > 1$ but $m \geq a$, the per-pass advantage is exhausted by the extra passes. Third, if $m < 1$ as well as $a > 1$, the forward-only method wins for two independent reasons; this is a stronger claim than either effect alone and should be reported separately.

[Conjecture] Early stopping and schedule design act primarily on m, not on a. A fair comparison should therefore report both the stopping rule and the realised pass count; comparing a forward-only method at its own convergence with a BP baseline at a fixed epoch budget confounds the two. The W1 BP-scheduled and budget-matched blocks are designed to separate these effects, but the conjecture that schedule improvements generalise across datasets remains to be tested.

4.5. Summary of the testable model

The model has one definitional identity, (3), one algebraic win condition, (5), and one overhead correction, (7). Its empirical content is the pair (a, m) measured at a fixed target. The W1 design measures a in the matched-data-pass protocol, measures m from the pass count to the pre-registered target, retains the segments needed for c_f and c_b, and records node and GPU identity so that hardware variation is visible rather than absorbed into the treatment effect. The design does not vary width, depth, or device generation; those remain conjectures for a later study.

5. Mono-Forward and the Cost Hypothesis

5.1. The Mono-Forward update

MF trains a stack of stages with a local objective. In the variant used here, each stage produces a representation and a local prediction; the local loss is applied at the stage, and the path between stages is detached for the purpose of the local update. The detach is not an implementation detail: it is the mechanism that removes the global backward graph. The W1 ablation includes a variant without the M0 stage and variants with joint, simultaneous, and sequential training of the stages, so that the cost of the mechanism can be separated from the cost of the architecture.

5.2. Why a per-pass advantage is plausible

A forward-only pass avoids the backward sweep and the activation retention it requires. It may also run at a lower power draw because it has fewer synchronisation points. These effects can make $a > 1$. They do not

guarantee that $m \leq a$: local objectives can converge more slowly, and a method that needs more passes can lose end to end even with a substantial per-pass advantage. Section 4 states the condition; the W1 family measures both sides of it.

5.3. The cost hypothesis

[Conjecture] The end-to-end advantage of MF is not a consequence of removing the backward pass alone. It is jointly determined by the per-pass energy ratio and the pass multiplier at the target accuracy. A convincing demonstration must report both, with the target and the stopping rule fixed in advance. This is the hypothesis that the W1 design is built to test.

6. Measuring Cost Fairly: Two Protocols

6.1. The matched-data-pass protocol

In the matched-data-pass protocol, every arm performs the same number of data passes and the comparison asks for the energy per pass and the accuracy reached at that budget. This protocol isolates the cost of the update rule. It is the protocol used in the PPAM 2026 paper and it is the correct protocol for the question "which algorithm is cheaper for the same amount of training data?" It is not an end-to-end protocol, because the arms need not reach the same accuracy.

6.2. The matched-accuracy protocol

In the matched-accuracy protocol, every arm trains until it first reaches the pre-registered target $A^\star$, and the comparison asks for cumulative energy and passes to that point. Runs that never reach $A^\star$ are censored, not assigned the energy of their last pass. This protocol answers the end-to-end question and is the basis of the win condition in Section 4.

6.3. Instrumentation boundary

Energy is measured at the device level with NVML, using the same start and stop events for every arm. The traces record node identity, GPU UUID, start time, per-stage boundaries, and the evaluation segments. The analysis integrates power over the training interval and reports the per-pass and cumulative values separately. [W1 PLACEHOLDER: insert the final instrumentation-boundary paragraph and the NVML sampling interval; the frozen protocol fixes the boundary, not the hardware generation.]

6.4. Tuning and stopping

The tuning budget, search space, sampler, and pruner are fixed per arm. The confirmatory runs use the selected configuration; the BP-bestval control uses BP's best validation-selected configuration rather than a default. The BP-scheduled control changes only the schedule and the early-stopping setting, so that its difference from BP-bestval is interpretable. The budget-matched controls use the Protocol B rule, whose per-stage budgets are computed from the core BP runs after those runs complete; the rule is frozen before the controls are run.

7. The W1 Pre-Registered Design

7.1. Questions and family

W1 asks whether MF's per-pass advantage survives an end-to-end comparison, and whether any advantage is robust to tuned BP, a scheduled BP control, and a budget-matched control. The family has five core arms (BP, MF-Joint, MF-Simul, MF-Seq, and MF-Seq-nom0), two datasets (CIFAR-10 and CIFAR-100), and twenty paired seeds (42–61), for 200 core runs. The BP-bestval block adds 40 runs. The BP-scheduled block uses 50 tuning trials per dataset followed by a confirmatory comparison. The Seq-Global block and the budget-matched Protocol B block are gated separately; neither is opened by this skeleton.

7.2. Endpoints and decision rule

The primary endpoint is the matched-accuracy end-to-end energy ratio at the pre-registered target. Secondary endpoints are the per-pass energy ratio, the pass multiplier, test accuracy at the matched budget, and the stage-wise energy decomposition. The family-wise decision uses the pre-registered Holm procedure over the frozen family. [W1 PLACEHOLDER: insert the exact endpoint definitions and the family list from PREREG-phase2.md; do not paraphrase the decision rule in the final manuscript.]

7.3. Controls and gates

The BP-bestval control asks whether the result survives a properly tuned BP baseline. The BP-scheduled control asks whether a longer, scheduled BP run closes the gap. The budget-matched Protocol B control asks whether MF's advantage survives when both methods receive the same pass budget derived

from the BP epochs. Seq-Global is a separate robustness block and remains gated until its scheduled date. The gating is part of the pre-registration, not an analysis choice.

7.4. Deviations and provenance

Every deviation is recorded in the dated amendment log before the affected analysis is read. The freeze commit, the analysis-code hash, the configuration manifest hash, and the job-account records are retained with the results. The frozen paths are checked against the freeze tag before each block. No W1 outcome is inspected during health monitoring.

8. Results

8.1. Matched-data-pass energy

[W1 PLACEHOLDER: populate the matched-data-pass energy table from the frozen analysis; report per-pass energy, accuracy at the matched budget, and the paired-seed interval. Do not compare arms on elapsed time alone.]

8.2. End-to-end energy at the target

[W1 PLACEHOLDER: populate the matched-accuracy energy table and the win-condition figure; report the measured pair (a, m) with uncertainty, the ratio $R = m/a$, and the pre-registered decision.]

8.3. Pass multiplier and convergence

[W1 PLACEHOLDER: report the pass counts to the target, censored runs, and the convergence curves; separate the schedule effect from the algorithm effect.]

8.4. Tuned BP and scheduled BP controls

[W1 PLACEHOLDER: report BP-bestval and BP-scheduled results against the same target; state whether the control changes the sign or the size of the effect.]

8.5. Budget-matched Protocol B

[W1 PLACEHOLDER: report the Protocol B controls only after the core BP block is complete and the budget rule has been applied; label every ex-M0 quantity as an estimate.]

8.6. Robustness and heterogeneity

[W1 PLACEHOLDER: report seed-level heterogeneity, node/GPU checks, and the pre-registered sensitivity analyses. Do not turn health-monitoring elapsed times into an outcome table.]

9. Discussion and Limitations

9.1. What the result would mean

A measured $R < 1$ with a compatible uncertainty interval would support the end-to-end claim at the tested scale. A measured $a > 1$ with $R \geq 1$ would show that the per-pass advantage is real but insufficient, which is a useful negative result rather than a failed experiment. A measured $a \leq 1$ would reject the per-pass premise for this implementation and would make the end-to-end question moot at this scale.

9.2. Threats to validity

The main threats are hardware heterogeneity, instrumentation error, tuning budget mismatch, and the fixed architecture. The design records node and GPU identity, retains raw traces, freezes the tuning budget, and pre-registers the target. It cannot eliminate node-to-node variation, and it does not vary width, depth, or device generation. Those are scope limitations, not post-hoc excuses.

9.3. Relation to the conference paper

The journal version does not replace the conference result; it re-frames it. The matched-data-pass result answers a per-pass question. The W1 result answers the end-to-end question. Both are reported, with the protocol named in every table and figure, so that the two claims cannot be conflated.

10. Conclusion

Removing the backward pass is a mechanism, not an end-to-end guarantee. The cost model in this paper makes the missing step explicit: a forward-only method wins only when its per-pass advantage exceeds its pass multiplier, after fixed costs and uncertainty are accounted for. The W1 family is a pre-registered test of that condition on a fixed A100 platform. [W1 PLACEHOLDER: insert the one-sentence W1 conclusion after the frozen analysis; do not preview it here.]

CRediT authorship contribution statement

[W1 PLACEHOLDER: confirm the final CRediT roles with both authors.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The frozen configurations, analysis code, table shells, and provenance records are available in the project repository. [W1 PLACEHOLDER: insert the repository DOI or archive identifier and the JPDC data statement after the W1 analysis.]

Funding

This work was supported by the National Science Centre (NCN), Poland, under grant OPUS-29, DEC-2025/57/B/ST6/04377. [W1 PLACEHOLDER: confirm the final funding statement and any additional institutional support.]

Acknowledgements

We thank the PLGrid Infrastructure and ACK Cyfronet AGH for computational resources on the Athena cluster under grant PLG/2025/018341 and allocation plgoncotherapy2-gpu-a100. [W1 PLACEHOLDER: confirm whether the class-account allocation plglscclass26-gpu-a100 may be acknowledged; do not include it until the permitted-use question is resolved.]

References

[] Gong, J., Li, B., Abdulla, W., 2025. Mono-Forward: Revisiting Forward-Forward through objective-locality decomposition. arXiv preprint arXiv:2501.09238 doi:10.48550/arXiv.2501.09238.

[] Hinton, G., 2022. The Forward-Forward algorithm: Some preliminary investigations. arXiv preprint arXiv:2212.13345 doi:10.48550/arXiv.2212.13345.

[] Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors. Nature 323, 533–536. doi:10.1038/323533a0.

[] Zhao, G., Wang, T., Jin, Y., Lang, C., Li, Y., Ling, H., 2025. The cascaded forward algorithm for neural network training. Pattern Recognition 161, 111292. doi:10.1016/j.patcog.2024.111292.

Appendix A. Protocol Deviations and Provenance

[W1 PLACEHOLDER: list every dated deviation from PREREG-phase2.md, the freeze tag, the analysis-code hash, and the configuration-manifest hash. This appendix must be generated from the amendment log, not from memory.]

Appendix B. Instrumentation and Measurement Error

[W1 PLACEHOLDER: describe the NVML sampling interval, integration rule, start/stop boundaries, missing-sample handling, and the stage-sum validation.]

Appendix C. Supplementary Tables

[W1 PLACEHOLDER: place the full per-seed tables and the table-shell schema here; the header-only shells in analysis/w1_table_shells/ are the starting point.]